River Crossing

Green Book 2.2: Logic Reasoning

Problem. Four people, A, B, C and D need to get across a river. The only way to cross the river is by an old bridge, which holds at most 2 people at a time. Being dark, they can't cross the bridge without a torch, of which they only have one. So each pair can only walk at the speed of the slower person. They need to get all of them across to the other side as quickly as possible. A is the slowest and takes 10 minutes to cross; B takes 5 minutes; C takes 2 minutes; and D takes 1 minute.

What is the minimum time to get all of them across to the other side?

Solution.

The key idea is to make the two slowest people, A and B, cross together. Since a pair moves at the speed of the slower person, this crossing costs only 10 minutes.

We use C and D to help return the torch:

$$\begin{array}{cc} C_D \rightarrow & 2 \\ D \leftarrow & 1 \\ A_B \rightarrow & 10 \\ C \leftarrow & 2 \\ C_D \rightarrow & 2 \end{array}$$

Thus the total time is

$$2 + 1 + 10 + 2 + 2 = 17.$$

Therefore, the minimum time is

$$\boxed{17}.$$

Comparison with greedy.

Greedy

$$\begin{array}{ccc} & A_D \rightarrow & A \\ & D \leftarrow & D \\ & B_D \rightarrow & B \\ & D \leftarrow & D \\ & C_D \rightarrow & C \\ \Longrightarrow & & A + B + C + 2D \end{array}$$

Optimal

$$\begin{array}{ccc} & C_D \rightarrow & C \\ & D \leftarrow & D \\ & A_B \rightarrow & A \\ & C \leftarrow & C \\ & C_D \rightarrow & C \\ \Longrightarrow & & A + 3C + D \end{array}$$

$$\boxed{\min(A + B + C + 2D,\ A + 3C + D) = A + C + D + \min(B + D,\ 2C)}$$

For this problem,

$$10 + 2 + 1 + \min(6, 4) = 17.$$

General case.

For $n > 4$, if A, B are the two slowest remaining and C, D are the two fastest, then one reduction step costs

$$\boxed{A + C + D + \min(B + D,\ 2C)}.$$

After this step, A and B are across permanently, while C and D are back with the torch. Hence the state reduces by

$$\boxed{n \to n - 2}.$$